

RAI-818: INTELLIGENT **MOBILE ROBOTICS**

Mobile Robot Kinematics – I

- **Introduction**
- **Kinematic Model of Wheeled Mobile Robot**
- **Generalized Wheel Model**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:

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Kinematics and Dynamics



- **Introduction**
 - **Mobile Robot Kinematics?**

Mobile Robot Kinematics?

- Kinematics is the *study* or the *mathematics* of motion *without considering the forces/efforts* that affect the motion
 - Deals with geometric relationships that govern the system
 - Deals with the relationship between control parameters and the behavior of a system in state space
- Needed to?
 - *To understand* the behavior of the system and *design* the system
 - *To design* suitable controllers, navigation system and adjust their performances
 - *To predict or estimate* the system parameters, to illustrate or mimic or simulate the real system, etc.



Defining futures

Kinematics and Dynamics



- **Introduction**
 - Mobile Robot Kinematics?
 - **Degrees of Freedom (DoF)**

Degrees of Freedom (DoF)

- Minimum number of variables (or number of independent variables) needed to describe the system
 - For the *land based wheeled mobile robots and water surface vehicles: degree of freedom is 3* (2x translational, 1x orientation in a plane)
 - For the *aerial, space and under water robots: degree of freedom is 6* (3x translational, 3x orientations in space)



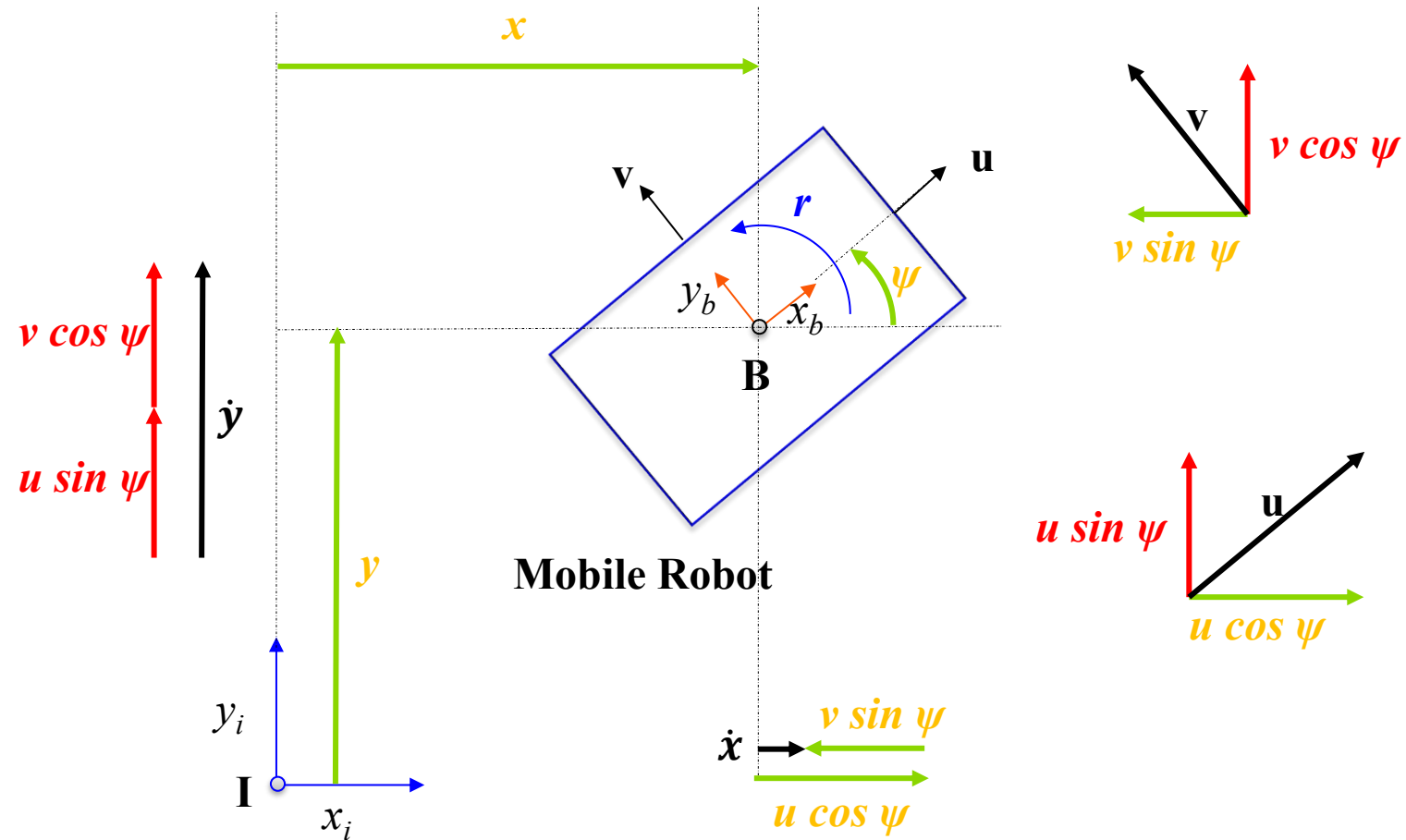
Defining futures

Kinematics and Dynamics



- Introduction
- **Kinematic Model of Mobile Robot**

Kinematic Model of Mobile Robot



Defining futures

Kinematics and Dynamics



- Introduction
- **Kinematic Model of Mobile Robot**

Kinematic Model of Mobile Robot

- x : Forward displacement of the mobile robot w.r.t. $\mathbf{I}$
- y : Lateral displacement of the mobile robot w.r.t. $\mathbf{I}$
- ψ : Angular displacement of the mobile robot w.r.t. $\mathbf{I}$
- u : Forward velocity of the mobile robot w.r.t. $\mathbf{I}$
- v : Lateral velocity of the mobile robot w.r.t. $\mathbf{I}$
- r : Angular velocity of the mobile robot w.r.t. $\mathbf{I}$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} u \cos \psi - v \sin \psi \\ u \sin \psi + v \cos \psi \\ r \end{bmatrix} \quad \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$\dot{\eta} = J(\dot{\psi})\zeta$ $J(\dot{\psi})$ is the Jacobian (or velocity transformation) matrix

- It the relation between the velocity input commands (ζ) and the derivatives of generalized coordinates ($\dot{\eta}$)



Defining futures

Kinematics and Dynamics



- Introduction
- **Kinematic Model of Mobile Robot**
 - Forward/Inverse Differential Kinematics

Forward/Inverse Differential Kinematics

Forward Differential Kinematics

- For given velocity input commands, find the derivatives of generalized coordinates (i.e., find the motion of the system).

$$\dot{\eta} = J(\dot{\psi})\zeta$$

Simulating or analyzing the system at the velocity level

Inverse Differential Kinematics

- For desired (given) derivatives of generalized coordinates (or a given position/trajectory) find corresponding velocity input commands

$$\zeta = J^{-1}(\dot{\eta})$$

Controlling the system at the velocity level



Defining futures

Kinematics and Dynamics



- Introduction
- **Kinematic Model of Mobile Robot**
 - Forward/Inverse Differential Kinematics
 - **Degree of Maneuverability**

Degree of Maneuverability

- Includes both the degree of freedom that the robot manipulates directly through wheel velocity and the degree of freedom that it indirectly manipulates by changing the steering configuration and moving.
- Total number of controllable inputs
- Degree of maneuverability depends on the kinematic configuration and actuator arrangement of the mobile robot.
- Sum of the degree of mobility and steerability
Maneuverability = Mobility + Steerability
$$\delta_M = \delta_m + \delta_s$$
- The known mobile robot kinematic mode is : $\dot{\eta} = J(\psi)\zeta$
- Based on wheel configuration: $\zeta = W\omega$



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Kinematics and Dynamics



- Introduction
- **Kinematic Model of Mobile Robot**
 - Forward/Inverse Differential Kinematics
 - Degree of Maneuverability
 - **Type of Wheels**

Type of Wheels

- ***Solid***
 - ***Conventional Wheels***
(Pure rolling - infinite lateral friction)
 - ***Fixed Wheel*** (Hub axis is fixed w.r.t. vertical axis)
 - ***Steerable Wheel*** (Hub axis can rotate about the vertical axis)
 - ***Non-Conventional***
 - ***Omni-Directional***
 - ***Mecanum Wheel***
- ***Inflated Wheel***



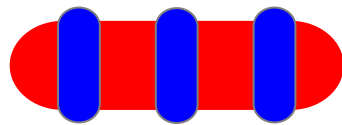
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Kinematics and Dynamics

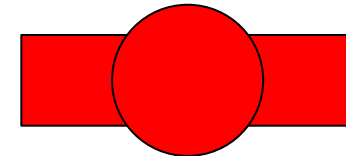
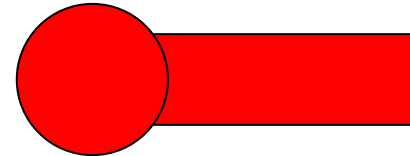
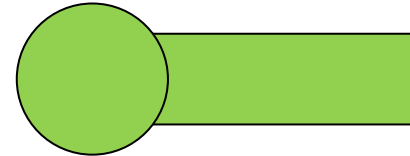
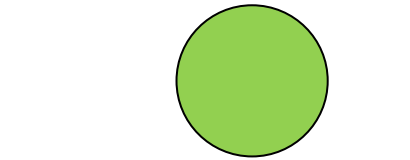


- Introduction
- **Kinematic Model of Mobile Robot**
 - Forward/Inverse Differential Kinematics
 - Degree of Maneuverability
 - **Type of Wheels**

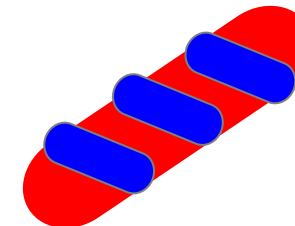
Type of Wheels



Fixed Wheel



Steerable Wheel



Mecanum Wheel



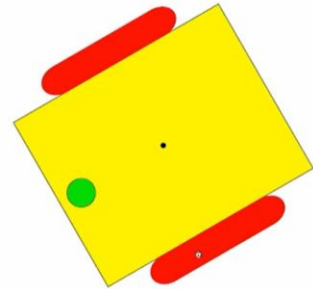
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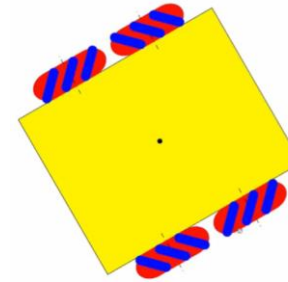


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 - Forward/Inverse Differential Kinematics
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 - **Type of Wheels**

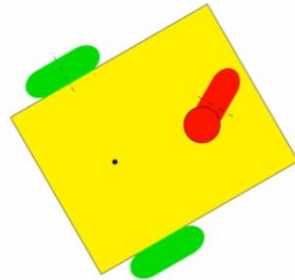
Type of Wheels



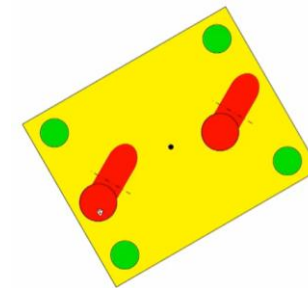
$$\begin{aligned}\delta_m &= 2 \\ \delta_s &= 0 \\ \delta_M &= 2\end{aligned}$$



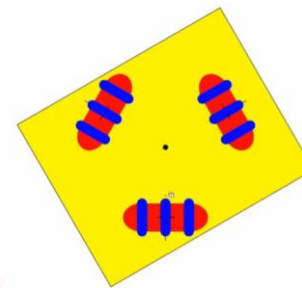
$$\begin{aligned}\delta_m &= 4 \\ \delta_s &= 0 \\ \delta_M &= 4\end{aligned}$$



$$\begin{aligned}\delta_m &= 1 \\ \delta_s &= 1 \\ \delta_M &= 2\end{aligned}$$



$$\begin{aligned}\delta_m &= 2 \\ \delta_s &= 2 \\ \delta_M &= 4\end{aligned}$$



$$\begin{aligned}\delta_m &= 3 \\ \delta_s &= 0 \\ \delta_M &= 3\end{aligned}$$



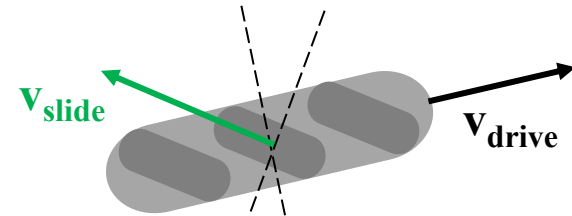
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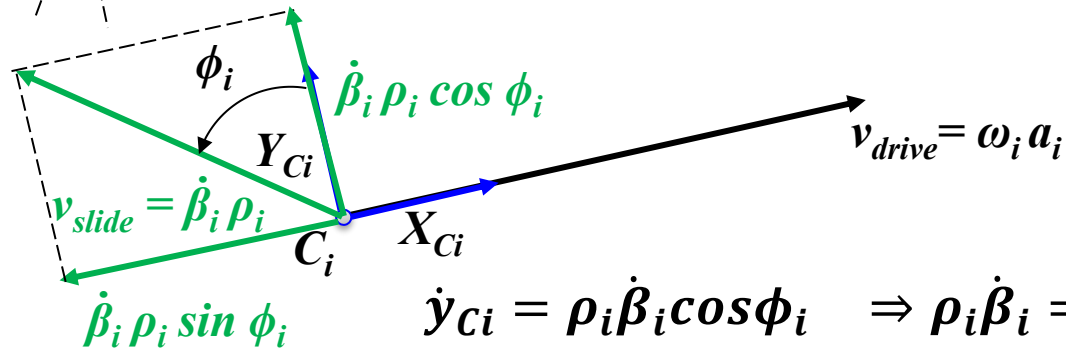


- Introduction
- Kinematic Model of Mobile Robot
- **Generalized Wheel Model**

Generalized Wheeled Model



$$\omega_i = \begin{bmatrix} \frac{1}{a_i} & \frac{1}{a_i} \tan \phi_i \end{bmatrix} \begin{bmatrix} \dot{x}_{Ci} \\ \dot{y}_{Ci} \end{bmatrix}$$



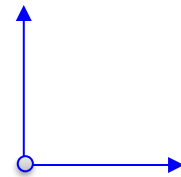
$$\dot{y}_{Ci} = \rho_i \dot{\beta}_i \cos \phi_i \Rightarrow \rho_i \dot{\beta}_i = \frac{\dot{y}_{Ci}}{\cos \phi_i}$$

$$\dot{x}_{Ci} = \omega_i a_i - \rho_i \dot{\beta}_i \sin \phi_i$$

$$\dot{x}_{Ci} = \omega_i a_i - \dot{y}_{Ci} \tan \phi_i$$

$$\omega_i a_i = \dot{x}_{Ci} + \dot{y}_{Ci} \tan \phi_i$$

$$\Rightarrow \omega_i = \frac{1}{a_i} (\dot{x}_{Ci} + \dot{y}_{Ci} \tan \phi_i)$$



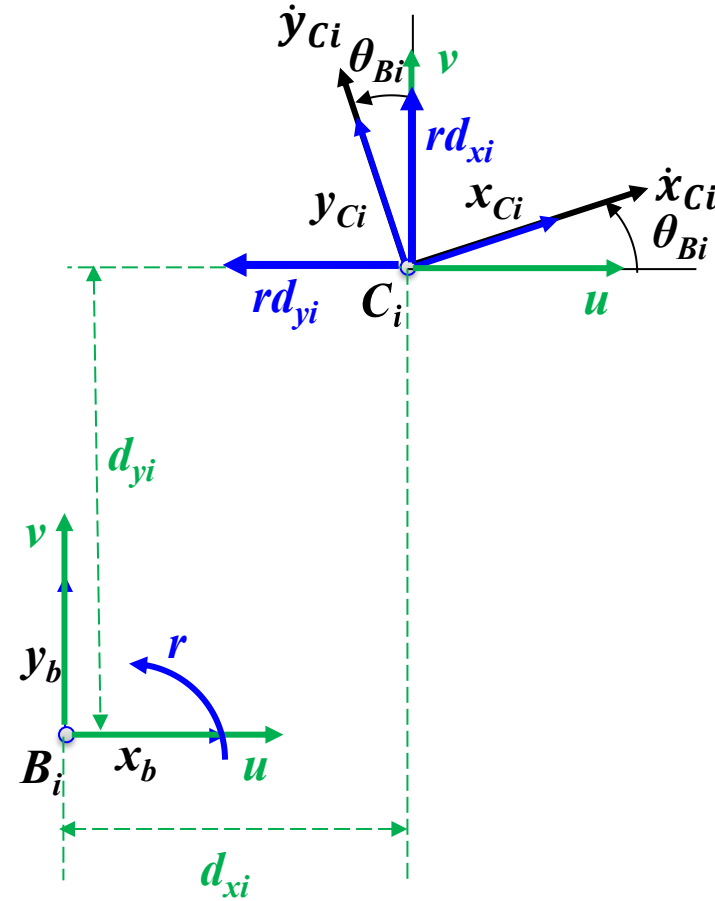
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Generalized Wheeled Model



$${}^{C_i}v_{C_i} = \begin{bmatrix} \dot{x}_{C_i} \\ \dot{y}_{C_i} \end{bmatrix}$$

$${}^B v_{C_i} = {}^{B_i}R(\theta_{B_i}) \begin{bmatrix} \dot{x}_{C_i} \\ \dot{y}_{C_i} \end{bmatrix}$$

$$v_{XB} = \dot{x}_{C_i} \cos \theta_{B_i} - \dot{y}_{C_i} \sin \theta_{B_i}$$

$$v_{YB} = \dot{x}_{C_i} \sin \theta_{B_i} + \dot{y}_{C_i} \cos \theta_{B_i}$$

$${}^B v_{C_i} = \begin{bmatrix} \cos \theta_{B_i} & -\sin \theta_{B_i} \\ \sin \theta_{B_i} & \cos \theta_{B_i} \end{bmatrix} \begin{bmatrix} \dot{x}_{C_i} \\ \dot{y}_{C_i} \end{bmatrix}$$



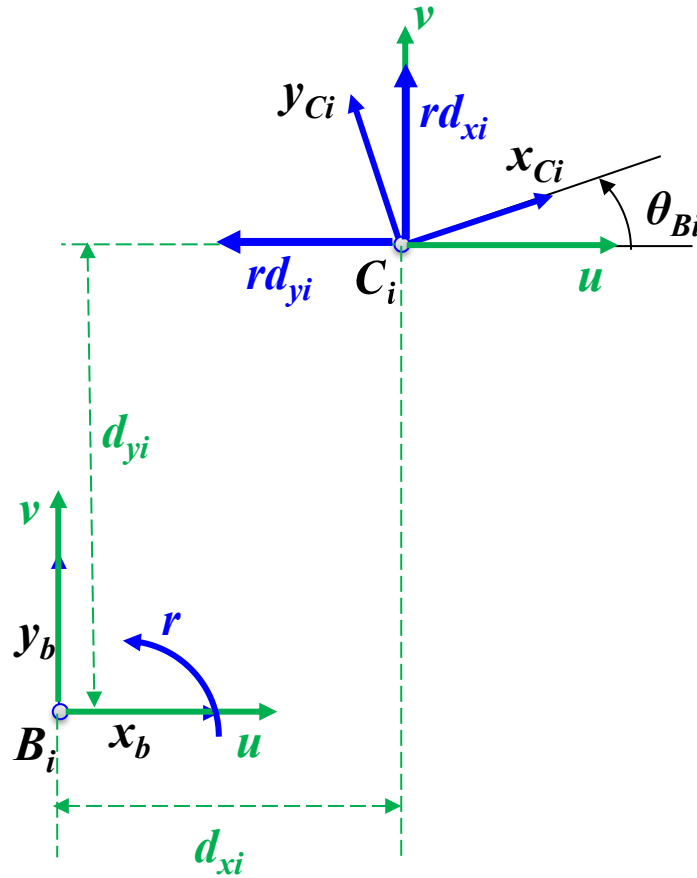
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Generalized Wheeled Model



$${}^B v_{Ci} = \begin{bmatrix} \cos\theta_{Bi} & -\sin\theta_{Bi} \\ \sin\theta_{Bi} & \cos\theta_{Bi} \end{bmatrix} \begin{bmatrix} \dot{x}_{Ci} \\ \dot{y}_{Ci} \end{bmatrix}$$

$$u - rd_{yi} = \dot{x}_{Ci} \cos\theta_{Bi} - \dot{y}_{Ci} \sin\theta_{Bi}$$

$$v + rd_{xi} = \dot{x}_{Ci} \sin\theta_{Bi} + \dot{y}_{Ci} \cos\theta_{Bi}$$

$${}^B v_{Ci} = \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \begin{bmatrix} \cos\theta_{Bi} & -\sin\theta_{Bi} \\ \sin\theta_{Bi} & \cos\theta_{Bi} \end{bmatrix} \begin{bmatrix} \dot{x}_{Ci} \\ \dot{y}_{Ci} \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_{Ci} \\ \dot{y}_{Ci} \end{bmatrix} = \begin{bmatrix} \cos\theta_{Bi} & \sin\theta_{Bi} \\ -\sin\theta_{Bi} & \cos\theta_{Bi} \end{bmatrix} \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$



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Generalized Wheeled Model

$$\omega_i = \begin{bmatrix} \frac{1}{a_i} & \frac{1}{a_i} \tan \phi_i \end{bmatrix} \begin{bmatrix} \dot{x}_{Ci} \\ \dot{y}_{Ci} \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_{Ci} \\ \dot{y}_{Ci} \end{bmatrix} = \begin{bmatrix} \cos \theta_{Bi} & \sin \theta_{Bi} \\ -\sin \theta_{Bi} & \cos \theta_{Bi} \end{bmatrix} \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$$\omega_i = \begin{bmatrix} \frac{1}{a_i} & \frac{1}{a_i} \tan \phi_i \end{bmatrix} \begin{bmatrix} \cos \theta_{Bi} & \sin \theta_{Bi} \\ -\sin \theta_{Bi} & \cos \theta_{Bi} \end{bmatrix} \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$$\omega_i = \begin{bmatrix} \frac{1}{a_i} & \frac{1}{a_i} \tan \phi_i \end{bmatrix} \begin{bmatrix} \cos \theta_{Bi} & \sin \theta_{Bi} \\ -\sin \theta_{Bi} & \cos \theta_{Bi} \end{bmatrix} \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \end{bmatrix}$$



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$$\omega_i = \begin{bmatrix} \frac{1}{a_i} & \frac{1}{a_i} \tan \phi_i \end{bmatrix} \begin{bmatrix} \cos \theta_{Bi} & \sin \theta_{Bi} \\ -\sin \theta_{Bi} & \cos \theta_{Bi} \end{bmatrix} \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

- ω_i : Angular velocity of the i^{th} wheel
- a_i : Radius of the i^{th} wheel
- θ_{Bi} : Angle between the vehicle frame B and the wheel frame (C_i)
- d_{xi} and d_{yi} : position coordinates of C_i w.r.t. B
- ϕ_i : Angle between the roller axis and the x_{C_i} axis
- u : Forward velocity of the mobile robot w.r.t. I
- v : Lateral velocity of the mobile robot w.r.t. I
- r : Angular velocity of the mobile robot w.r.t. I



Defining futures

Kinematics and Dynamics



- Introduction
- Kinematic Model of Mobile Robot
- **Generalized Wheel Model**
 - **Differential Drive Robot**

Differential Drive Robot

$$\omega_i = \begin{bmatrix} \frac{1}{a_i} & \frac{1}{a_i} \tan \phi_i \end{bmatrix} \begin{bmatrix} \cos \theta_{Bi} & \sin \theta_{Bi} \\ -\sin \theta_{Bi} & \cos \theta_{Bi} \end{bmatrix} \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$$\omega_1 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -d \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u - rd)$$

$$\omega_2 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & d \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u + rd)$$

$$\begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{a} & -\frac{d}{a} \\ \frac{1}{a} & \frac{d}{a} \end{bmatrix} \begin{bmatrix} u \\ r \end{bmatrix}$$

$$\begin{bmatrix} u \\ v \\ r \end{bmatrix} = \begin{bmatrix} \frac{a}{2} & \frac{a}{2} \\ 0 & 0 \\ -\frac{a}{2d} & \frac{a}{2d} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}$$

$$\begin{bmatrix} u \\ r \end{bmatrix} = \begin{bmatrix} \frac{a}{2} & \frac{a}{2} \\ -\frac{a}{2d} & \frac{a}{2d} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}$$

$$\zeta = W\omega$$

$$d_{y1} = d$$

$$d_{x1} = 0$$

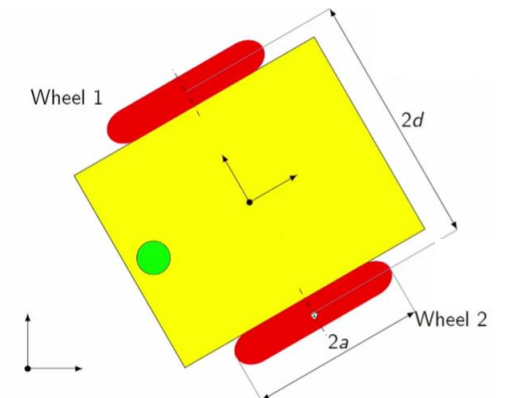
$$a_1 = a$$

$$d_{y2} = -d$$

$$d_{x2} = 0$$

$$a_2 = a$$

$$\phi_1 = \phi_2 = 0$$



Defining futures

Kinematics and Dynamics



- Introduction
- Kinematic Model of Mobile Robot
- **Generalized Wheel Model**
 - Differential Drive Robot
 - **Skid Steer Robot**

Skid Steer Robot

$$d_{x1} = l \quad d_{x2} = -l \quad d_{x3} = -l \quad d_{x4} = l \quad d_{y1} = d \quad d_{y2} = d \quad d_{y3} = -d \quad d_{y4} = -d$$

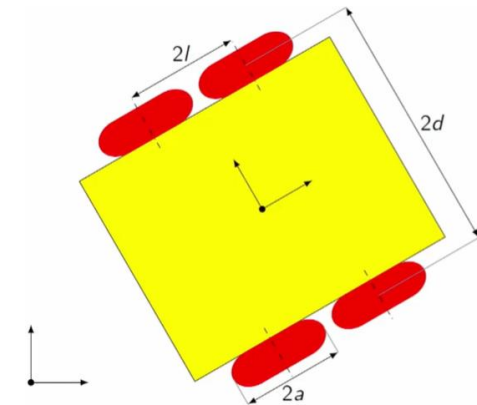
$$a_1 = a_2 = a_3 = a_4 = a \quad \theta_{B1} = \theta_{B2} = \theta_{B3} = \theta_{B4} = 0 \quad \phi_1 = \phi_2 = \phi_3 = \phi_4 = 0$$

$$\omega_1 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -d \\ l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u - rd)$$

$$\omega_2 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -d \\ -l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u - rd)$$

$$\omega_3 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} d \\ -l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u + rd)$$

$$\omega_4 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} d \\ l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u + rd)$$



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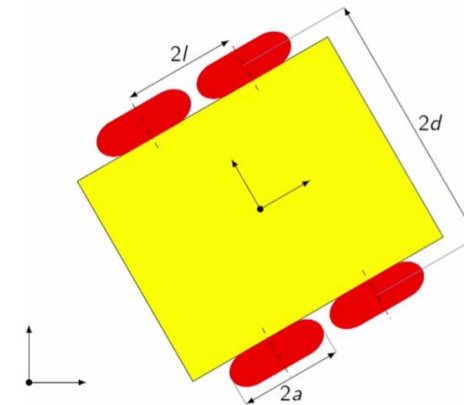
Differential Drive Robot

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{a} & -\frac{d}{a} \\ \frac{1}{a} & -\frac{d}{a} \\ 1 & d \\ \frac{1}{a} & \frac{d}{a} \end{bmatrix} \begin{bmatrix} u \\ r \end{bmatrix}$$

$$\begin{bmatrix} u \\ r \end{bmatrix} = \begin{bmatrix} \frac{a}{4} & \frac{a}{4} & \frac{a}{4} & \frac{a}{4} \\ -\frac{a}{4d} & -\frac{a}{4d} & \frac{a}{4d} & \frac{a}{4d} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix}$$

$$\begin{bmatrix} u \\ v \\ r \end{bmatrix} = \begin{bmatrix} \frac{a}{4} & \frac{a}{4} & \frac{a}{4} & \frac{a}{4} \\ 0 & 0 & 0 & 0 \\ -\frac{a}{4d} & -\frac{a}{4d} & \frac{a}{4d} & \frac{a}{4d} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix}$$

$$\zeta = W\omega$$



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 - **Omni-Drive Robot**

Omni-Drive Robot

$$d_{x1} = l \cos(0) \quad d_{x2} = l \cos(120) \quad d_{x3} = l \cos(240) \quad \phi_1 = \phi_2 = \phi_3 = 0$$

$$d_{y1} = l \sin(0) \quad d_{y2} = l \sin(120) \quad d_{y3} = l \sin(240) \quad a_1 = a_2 = a_3 = a$$

$$\theta_{B1} = 90 \quad \theta_{B2} = 210 \quad \theta_{B3} = 330$$

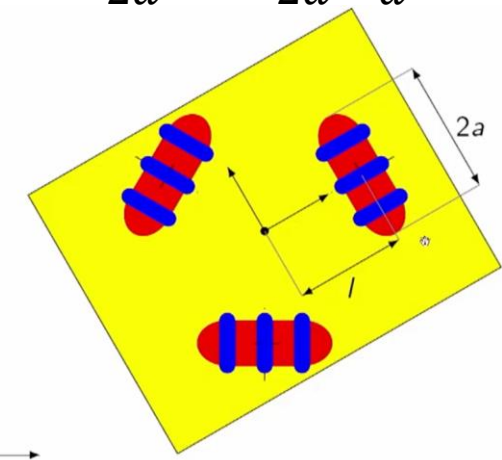
$$\omega_1 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (v + lr)$$

$$\omega_2 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & -\frac{\sqrt{3}l}{2} \\ 0 & 1 & -\frac{l}{2} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{2a} (-\sqrt{3}u - v + 2lr)$$

$$\omega_3 = \begin{bmatrix} \frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & -\frac{\sqrt{3}l}{2} \\ 0 & 1 & -\frac{l}{2} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{2a} (\sqrt{3}u - v + 2lr)$$

$$\Rightarrow \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \zeta = \begin{bmatrix} 0 & -\frac{a\sqrt{3}}{3} & \frac{a\sqrt{3}}{3} \\ \frac{2a}{3} & \frac{a}{3} & -\frac{a}{3} \\ \frac{a}{3l} & \frac{a}{3l} & \frac{a}{3l} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = W\omega$$

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{a} & \frac{l}{a} \\ -\frac{\sqrt{3}}{2a} & -\frac{1}{2a} & \frac{l}{a} \\ \frac{\sqrt{3}}{2a} & -\frac{1}{2a} & \frac{l}{a} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$



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 - **Mecanum Wheel Drive Robot**

Mecanum Wheel Drive Robot

$$d_{x1}=l \quad d_{x2}=-l \quad d_{x3}=-l \quad d_{x4}=l \quad d_{y1}=d \quad d_{y2}=d \quad d_{y3}=-d \quad d_{y4}=-d$$

$$\theta_{B1} = \theta_{B2} = \theta_{B3} = \theta_{B4} = 0 \quad a_1 = a_2 = a_3 = a_4 = a \quad \phi_1 = 45, \phi_2 = -45, \phi_3 = 45, \phi_4 = -45,$$

$$\omega_1 = \begin{bmatrix} \frac{1}{a} & \frac{1}{a} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -d \\ 0 & 1 & l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u + v - r[d - l])$$

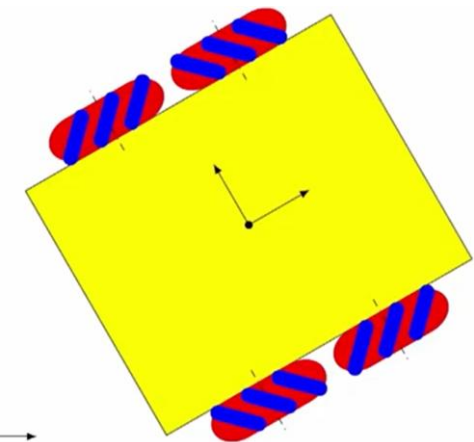
$$\omega_2 = \begin{bmatrix} \frac{1}{a} & \frac{1}{a} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -d \\ 0 & 1 & -l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u - v - r[d - l])$$

$$\omega_3 = \begin{bmatrix} \frac{1}{a} & \frac{1}{a} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & d \\ 0 & 1 & -l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u + v + r[d - l])$$

$$\omega_4 = \begin{bmatrix} \frac{1}{a} & \frac{1}{a} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & d \\ 0 & 1 & l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \frac{1}{a} (u - v + r[d - l])$$

$$\Rightarrow \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \zeta = \frac{a}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -\frac{1}{d-l} & -\frac{1}{d-l} & \frac{1}{d-l} & \frac{1}{d-l} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} = W\omega$$

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \frac{1}{a} \begin{bmatrix} 1 & 1 & -d+l \\ 1 & -1 & -d+l \\ 1 & 1 & d-l \\ 1 & -1 & d-l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$



Defining futures

Kinematics and Dynamics



- Introduction
- Kinematic Model of Mobile Robot
- **Generalized Wheel Model**
 - Differential Drive Robot
 - Skid Steer Robot
 - Omni-Drive Robot
 - Mecanum Wheel Drive Robot
 - **Tricycle Wheel Drive Robot**

Tricycle Wheel Drive Robot

$$d_{xI} = l \quad d_{yI} = 0 \quad a_I = a \quad \theta_{BI} = \theta_I \quad \phi_I = 0$$

$$\omega_i = \begin{bmatrix} \frac{1}{a_i} & \frac{1}{a_i} \tan \phi_i \end{bmatrix} \begin{bmatrix} \cos \theta_{Bi} & \sin \theta_{Bi} \\ -\sin \theta_{Bi} & \cos \theta_{Bi} \end{bmatrix} \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}$$

$$\begin{bmatrix} v_{1x} \\ v_{1y} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -d_{yi} \\ 0 & 1 & d_{xi} \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & l \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \begin{bmatrix} u \\ v + lr \end{bmatrix}$$

$$\begin{bmatrix} v_{1x} \\ v_{1y} \end{bmatrix} = \begin{bmatrix} \omega_1 a_1 \cos \theta_1 \\ \omega_1 a_1 \sin \theta_1 \end{bmatrix} = \begin{bmatrix} u \\ v + lr \end{bmatrix}$$

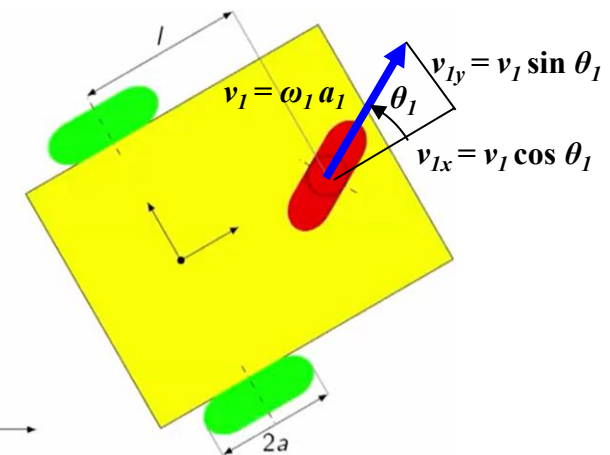
As per the pure rolling condition and based on wheel configuration, $v = 0$

$$v_{Iy} = \omega_I a_I \cos \theta_I = u$$

$$v_{Iy} = \omega_I a_I \sin \theta_I = v + lr = lr$$

$$\omega_1 = \frac{1}{a_1} \sqrt{u^2 + l^2 r^2}$$

$$\theta_1 = \tan^{-1} \left(\frac{lr}{u} \right)$$



Defining futures